Flame Tests: Identifying Metal Ions by Color

U2 / S2.5

BEFORE THIS IS COPIED — OPEN ITEMS

- **Draft.** A flame-test lab at S2.5 is not yet a confirmed lab placement in courses/chemistry/DECISIONS.md. It becomes final once you confirm it.
- **Every hazard and disposal statement** on the handout is marked not yet verified below. Check each against the SDS and district policy and date it.

PREP — THE DAY BEFORE

WHEN	TASK	DETAIL
Day before	Check stock solutions	Use the solutions on your list. Concentrations are not recorded yet; write them here so the next batch matches: _____. Each beaker needs enough solution to cover the bottom few centimetres of a stick.
Day before	Make the unknowns	Pour a beaker of each and label only the letter: **A = potassium nitrate · B = copper(II) nitrate · C = strontium chloride** (your choice of potassium, copper and strontium). A is potassium, not copper, so no group tests known copper at station 4 and then a copper unknown straight after it.
Before class	Soak the sticks	Stand wooden stir sticks in each of the twelve beakers, one per student in the period, so each class arrives to soaked sticks. Reload each beaker between periods.
Morning of	Load stations	Six stations, each with a burner, a striker and two labeled beakers of soaking sticks (layout below). Sticks per period = students × 12.
Before 1st period	Room	Clear the sinks so used sticks can drop in. Notebooks stay closed and away from flames except when recording.
Day before	Boric acid (demo only)	Make about 100 mL of roughly 4% boric acid in water: steps in the demo section below. Teacher only; students do not handle it.

Station layout

STATION	SAMPLE 1	SAMPLE 2
1	Lithium chloride, $LiCl$	Sodium nitrate, $NaNO_3$
2	Potassium nitrate, KNO_3	Calcium nitrate, $Ca(NO_3)_2$
3	Strontium chloride, $SrCl_2$	Barium nitrate, $Ba(NO_3)_2$
4	Copper(II) nitrate, $Cu(NO_3)_2$	Magnesium chloride, $MgCl_2$
5	Iron(III) nitrate, $Fe(NO_3)_3$	Unknown A — potassium nitrate
6	Unknown B — copper(II) nitrate	Unknown C — strontium chloride

TIMING — 50-MINUTE PERIOD

MIN	PHASE	WATCH FOR
0–5	Pre-lab check, goggles on, hair tied	No pre-lab, no burner.
5–8	Part A: light burners at the first station	Yellow, smoky flames: open the air.
8–38	Part B and C: six rotations of 5 minutes	Call "rotate" on the minute. Groups that finish early wait; they do not drift.
38–40	Boric acid demo at the front	Lights low if you can; green shows best in the first seconds.
40–45	Finish the table, compare with the group	Colors written from memory.

45–50	Cleanup, goggles off last	Sticks in the trash instead of the sink.

FAILURE POINTS

- **Everything looks yellow.** Sodium is everywhere, and the burning wood is yellow-orange. Say: "Record the first two seconds, then the stick goes in the sink."
- **Potassium's lilac is faint.** Any trace of sodium swamps it. Students who see yellow at station 2 or for unknown A usually put a stick back in a beaker or touched the tip.
- **Sticks put back in a beaker.** One used stick contaminates the beaker and every stick soaking in it, for every later group. Watch the first rotation closely; if it happens, swap the beaker and its sticks.
- **The stick catches fire.** Usually from holding it in the flame too long. Tip out, into the sink, new stick.
- **Magnesium "looks orange."** With no color of its own, students record the stick or a trace of sodium. Say: "If you saw nothing but the stick burning, that is the result."
- **"Red" for three compounds.** Push for crimson, bright red or orange-red. Lithium, strontium and calcium are where students lose the unknown.

DEMO — BORIC ACID, TEACHER ONLY

Water solution only. Never methanol, not even 50% methanol: the mixture is still flammable, its flame is hard to see, and an open container of it near a burner is the setup of the "rainbow" demonstration flash fires. Check the boric acid SDS first: many list it as a reproductive toxicant (Category 1B), so its pregnancy guidance decides who makes it up and runs the demo.

Prep, about 4%, 100 mL

1. Goggles and nitrile gloves. Weigh away from drafts and pour the powder gently; the dust is the main exposure.
2. Weigh 4 g of boric acid, H_3BO_3, into a 250 mL beaker.
3. Add about 80 mL of warm, not boiling, distilled water and stir with a glass rod until clear.
4. Cool to room temperature and bring to 100 mL. Roughly 5 g dissolves per 100 mL at room temperature, so 4 g stays in solution; if crystals form, add a little water.
5. Label: boric acid, H_3BO_3, about 4% (aq), date, initials, SDS hazard. Soak a few sticks in it before class, like the station beakers.

Running it

1. At the front, after the last rotation, keep the beaker well away from the burner and hold one soaked stick in the flame the same way students did.
2. Ask: "Green, but not barium's pale green, and boron is not a metal. What is still happening to its electrons?" Same answer as every station: excited electrons dropping back and giving off photons.
3. Used stick in the sink with the others.

EXPECTED RESULTS

A good group records the reference color for all seven ions on the table and names all three unknowns. A group that records "red" without more detail still gets A and B, but cannot separate C from lithium chloride: that is exactly post-lab question 2.

SAMPLE	EXPECTED (S2.5 NOTES)	WHAT STUDENTS OFTEN WRITE
$LiCl \cdot SrCl_2 \cdot Ca(NO_3)_2$	Crimson red · red · orange-red	"Red" for all three
$NaNO_3$	Yellow	Yellow, strong and long-lasting
KNO_3	Lilac	Faint purple, or yellow if contaminated
$Ba(NO_3)_2 \cdot Cu(NO_3)_2$	Pale green · blue-green	"Green" for both; copper is the brighter, bluer one

Magnesium chloride, $MgCl_2$	Little or no added color; most of its emission is ultraviolet	"Yellow-orange": that is the stick or sodium, not magnesium
Boric acid, H_3BO_3 (demo)	Green, brightest in the first seconds	Compared with barium's pale green; boric acid's is stronger
Iron(III) nitrate, $Fe(NO_3)_3$	Not on the reference table	Record what your first class sees and add it here: _____

SAFETY + DISPOSAL — NOT YET VERIFIED

- Nitrates as oxidizers (Na, K, Ca, Ba, Cu, Fe nitrate) — check each SDS. Verified on: _____
- Barium and copper compounds harmful by swallowing and on skin or eyes — check each SDS. Verified on: _____
- Magnesium chloride — check the SDS for its eye and skin statements. Verified on: _____
- Strontium chloride and calcium nitrate eye hazard — check each SDS. Verified on: _____
- Eyewash duration (15 minutes) — matches the room's standing procedure? Verified on: _____
- Disposal: students pour nothing out; the beakers are collected and the solution stored. The end-of-year disposal route for the copper, barium and strontium stock solutions follows the district chemical-waste policy — confirm it. Verified on: _____
- Used sticks (carrying traces of every solution, including barium): collected from the sink after class, drained, and put in the trash; confirm this is allowed. Verified on: _____
- Spills: students step back and report, and do not wipe up a solution themselves; confirm this matches the room's spill procedure. Verified on: _____
- Boric acid: SDS hazard statements, including reproductive toxicity, and who may handle it. Verified on: _____
- Boric acid leftover solution: disposal route under the district chemical-waste policy. Verified on: _____
- Alconox cleanup block is the standing default and needs no per-lab check.

PRE-LAB KEY

1. Ground and excited state. The electron absorbs energy from the flame and jumps from its ground state to a higher level (excited state). It drops back and gives off the energy as a photon of light. Both terms must appear.
2. Same color, $NaNO_3$ and NaCl. The metal ion (sodium) makes the color, not the nitrate or chloride part.
3. Hardest pair. Lithium (crimson red) and strontium (red); calcium (orange-red) is also accepted with one of them. An unknown red could be either, so color alone cannot decide.
4. Magnesium, no color. The electrons are still excited and still give off photons when they drop, but the drops are too big: most of the light is ultraviolet, which our eyes cannot detect. Needs both halves, plus "ultraviolet." "Magnesium has no electrons that move" earns nothing.
5. Part B, steps 1–2. A soaked stick from the station's beaker, held by the dry end; the wet tip goes at the edge of the flame, just above the blue inner cone.
6. Part B, step 3. After two or three seconds the wood burns yellow-orange, which is the stick's color, not the solution's.
7. Part B, steps 4–5. It goes in the sink. A stick put back into a beaker contaminates that solution and its soaking sticks for every later group.
8. Stick on fire. Take it out of the flame and drop it in the sink; never on the bench or the trash. Full credit needs the eyewash location written correctly.
9. Data table. Twelve rows in station order, three columns, in ink; expected colors filled from the reference table, magnesium from the Background, iron and the unknowns left blank.

POST-LAB KEY

1. Identify A, B, C. A: potassium nitrate, lilac. B: copper(II) nitrate, blue-green. C: strontium chloride, red. Each needs the color from the student's own table. If a student names lithium chloride for C and backs it with a crimson-red observation, give partial credit and expect question 2 to catch it.
2. Unknown with two matches. C (red) could be strontium chloride or lithium chloride. The two reds are too close to separate by eye, so the color is a clue, not proof. Accept another test or comparing C side by side with both known samples. Calcium nitrate is also accepted as a second match.
3. Magnesium chloride. Little or no color beyond the stick. Its electrons are excited and drop back, but most of the photons are ultraviolet, shorter than about 400 nm, outside the visible range. Full credit names ultraviolet and says the drops still happen. Accept a faint color if the student says it was faint and still explains the ultraviolet.
4. A mismatch, with a cause. Accept a real procedural cause tied to what they saw: recording after the wood started burning (yellow-orange), a stick put back in a beaker or a touched tip bringing in sodium (yellow over lilac), holding the tip too low or too high in the flame (faint color). For iron, describing the color they saw and saying it is not on the reference table is enough. "Human error" earns nothing.

GRADING

Suggested split, PROVISIONAL until you set it: pre-lab 9 (1 each) · data table 12 (1 per row with a specific color) · post-lab 8 (2 each). Carbonless copy is graded individually.

MISSED LAB

If the setup is already torn down, the student completes a Missed Lab Analysis rather than the class re-running the lab. Hand them a real class data table from this period (all twelve rows, unknowns labeled A to C only) and have them answer the pre-lab questions 1 to 4 and all four post-lab questions from it.